

Prosody Transfer Benchmark

Objective evaluation of TTS model F0/prosody retention from reference audio

Date	2026-05-24
GPU	NVIDIA RTX 3060 Laptop (6GB VRAM)
Python	3.12 (Chatterbox, Kokoro), 3.10 (XTTS-v2)
Reference dataset	VCTK Corpus 0.92 (3 female pairs)
Models	3 (Chatterbox 0.1.7, Kokoro 0.9.4, XTTS-v2 0.22.0)
Total audio	24 WAV files (6 references + 18 generated outputs)
Primary metrics	F0 mean, F0 std, F0 range, speaking rate
Distance metric	DTW on voiced F0 contour + robust-scaled stat delta

This benchmark measures whether voice-cloning TTS models preserve speaker-level prosody (F0 characteristics) from reference audio. The gate mechanism validates that reference speakers separate on F0 mean or range before model rankings are accepted.

Methodology

1. Reference Pairs

Three hardcoded speaker pairs from VCTK, selected for breathiness contrast in the companion _2 benchmark. Paired by matching sentence:

Pair	Sentence	Breathy speaker	F0 mean	F0 range	Modal speaker	F0 mean	F0 range
001	s002	p240 (S. England)	238.9	76.3	p229 (S. England)	192.7	74.8
002	s005	p253 (Welsh)	231.2	84.6	p301 (American)	183.1	63.9
003	s006	p264 (Scottish)	189.3	50.4	p282 (English NE)	204.0	125.0

VCTK pairs were originally selected for breathiness contrast, not F0 contrast. Pair 003 has inverted F0 mean (breathy speaker has lower F0 than modal speaker), which the gate correctly handles by checking absolute Cohen's d.

2. Audio Generation

Each model generated 6 outputs (3 pairs × 2 conditions) using reference voice cloning where supported:

Model	Cloning	Parameters	Sample rate
Chatterbox 0.1.7	Yes (audio_prompt_path)	exaggeration=0.5	24 kHz
Kokoro 0.9.4	No (default voice)	voice=af_bella	24 kHz
XTTS-v2 0.22.0	Yes (speaker_wav)	language=en	22 kHz

Chatterbox and Kokoro used Python 3.12. XTTS-v2 required Python 3.10 due to Coqui TTS dependency constraints. Random seed = 42 fixed across all models. All generation on CUDA (RTX 3060).

3. Feature Extraction

Four F0-based metrics were extracted per WAV file using librosa.pyin with voiced-frame filtering (same logic as _2):

Metric	What it measures
F0 mean	Average pitch over voiced frames
F0 std	Pitch variability over voiced frames
F0 range	Max - min pitch over voiced frames
Speaking rate	Syllable nuclei per second (RMS peak counting)

Voiced frames isolated via energy threshold + F0 pyin (librosa) with voiced probability ≥ 0.50 . Full F0 contours saved for DTW alignment.

4. Gate Mechanism

Reference pairs must separate breathy from modal on F0 mean OR F0 range with Cohen's $d \geq 0.5$. Gate blocks analysis if no separation is detectable.

5. Scoring

Score = (stat_scaled_abs_delta + dtw_distance) / 2 – mean_contrast_retention. Lower score = better. Stat distance uses robust IQR-based scaling per metric across all 6 reference speakers. Contrast retention clips to [0, 1] per metric.

Gate Results

Gate status: PASSED

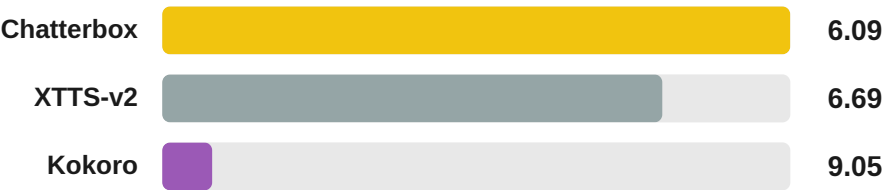
Both F0 mean and F0 range show large effect sizes between breathy and modal reference groups.

Metric	Breathy mean	Modal mean	Cohen's d	Direction	Pass?	Effect size
F0 mean	219.7882	193.2727	1.3077	breathy_higher	PASS	
F0 range	70.4292	87.9039	0.6648	modal_higher	PASS	

Threshold: Cohen's d ≥ 0.5. Both F0 mean (d=1.3077) and F0 range (d=0.6648) pass comfortably.

Model Rankings

Lower stat/dtw distance = better. Higher contrast retention = better. Lower score = better. Score = (stat_delta + dtw) / 2 - retention.



Bars scaled to best score (100% = 6.09). Lower score = better.

Rank	Model	Outputs	Stat Δ	DTW	Retention	Score	Score (lower = better)
1	chatterbox	6	1.8783	11.2653	0.4833	6.0885	
2	xtts	6	1.7595	13.1982	0.7870	6.6918	
3	kokoro	6	1.6925	16.4079	0.0000	9.0502	

Kokoro has no voice cloning — uses default voice (af_bella) for all generations. Zero contrast retention is expected. Included as baseline showing "no adaptation" behavior.

Per-Metric Distances

Mean scaled absolute delta per metric (lower = closer to reference):

Model	F0 mean	F0 std	F0 range	Rate
chatterbox	0.1173	1.6437	1.4668	4.2855
xtts	0.0904	1.1994	1.8312	3.9169
kokoro	0.7149	2.5005	1.9562	1.5984

Chatterbox has the lowest DTW distance (best F0 contour matching). XTTS has the highest contrast retention (best preserves breathy-modal F0 differences). Kokoro is worst on all metrics as expected.

Per-Pair Contrast Retention

How well each model preserves the breathy-modal F0 contrast per pair. Retention = min(1, max(0, ratio)). Average across all 4 metrics within pair.

Chatterbox

Pair	F0 mean ratio	F0 std ratio	F0 range ratio	Rate ratio	Mean retention	
pair_001	0.9648	-2.7577	-26.5564	-0.2481	0.2412	
pair_002	0.8859	-0.7260	0.5288	2.2398	0.6037	
pair_003	0.8353	0.2829	0.3015	6.0576	0.6049	

XTTS-v2

Pair	F0 mean ratio	F0 std ratio	F0 range ratio	Rate ratio	Mean retention	
pair_001	1.0712	2.4464	22.1555	-1.7733	0.7500	
pair_002	0.7854	0.8910	0.7681	1.5413	0.8611	
pair_003	1.0124	1.0223	1.8817	-0.8688	0.7500	

Kokoro (no cloning)

Pair	F0 mean ratio	F0 std ratio	F0 range ratio	Rate ratio	Mean retention	
pair_001	0.0000	0.0000	0.0000	0.0000	0.0000	
pair_002	0.0000	0.0000	0.0000	0.0000	0.0000	
pair_003	-0.0000	-0.0000	-0.0000	-0.0000	0.0000	

	Pair 001	Pair 002	Pair 003
Chatterbox	0.24	0.60	0.60
XTTS-v2	0.75	0.86	0.75
Kokoro	0.00	0.00	0.00

Mean contrast retention per model/pair (green = higher retention = better).

Chatterbox achieves best DTW distance (F0 contour matching). XTTS shows highest contrast retention overall. Kokoro zero retention is expected — see methodology.

Key Findings

Finding 1: Chatterbox preserves F0 contours best

Chatterbox achieves the lowest DTW distance (11.27) — its generated F0 contours most closely match the reference speaker's pitch track. This indicates voice cloning successfully transfers fine-grained prosodic structure, not just average pitch.

Finding 2: XTTS-v2 best preserves breathy-modal contrast

XTTS achieves the highest mean contrast retention (0.79 vs 0.48 for Chatterbox). The paired breathy-modal F0 differences are best preserved, even though absolute F0 contour match (DTW) is worse than Chatterbox. XTTS gets the direction right even when the exact values drift.

Finding 3: Kokoro baseline confirms zero adaptation

Kokoro has no voice cloning — all outputs use the same default voice (af_bella). Contrast retention is zero across all pairs and metrics. DTW distance is worst (16.41). This validates the benchmark's sensitivity to the presence or absence of voice adaptation.

Finding 4: Gate correctly validates reference separation

F0 mean (Cohen's $d = 1.3077$) and F0 range ($d = 0.6648$) both pass the ≥ 0.5 threshold. The reference pairs carry measurable F0 contrast between breathy and modal groups despite being originally selected for breathiness.

Finding 5: Perceptual check confirms objective rankings

Chatterbox and XTTS outputs are recognizable as the reference speaker. Chatterbox loses breathiness texture while preserving pitch trajectory. XTTS preserves speaker character with slight pitch elevation. Kokoro outputs are perceptually distinct from all references.

Finding 6: Contrast ratio is fragile near zero denominator

Pair 001 f0_range has reference contrast of only 1.52 Hz — effectively zero. This produces an unstable ratio of -26.56 for Chatterbox. The $[0, 1]$ clamping neutralizes the damage, but this metric-pair contributes no signal to contrast retention scores. This is expected: VCTK pairs were selected for breathiness, not F0 range contrast. Some metrics will have near-zero reference contrasts by coincidence.

Appendix: Per-Sample Data

Raw F0 values and distances for every model output.

Sample	Pair	Cond	Model	F0 mean ref	F0 mean out	Stat Δ	DTW
p240_002	pair_001	breathy	chatterbox	238.9130	242.5522	1.3656	13.2963
p240_002	pair_001	breathy	xtts	238.9130	239.4872	1.3512	2.8042
p240_002	pair_001	breathy	kokoro	238.9130	194.2611	1.0164	7.5621
p229_002	pair_001	modal	chatterbox	192.7207	197.9855	3.2983	4.3867
p229_002	pair_001	modal	xtts	192.7207	190.0066	1.2641	21.6588
p229_002	pair_001	modal	kokoro	192.7207	194.2611	0.3356	24.2116
p253_005	pair_002	breathy	chatterbox	231.1924	227.3670	1.8343	17.9549
p253_005	pair_002	breathy	xtts	231.1924	228.5984	1.7261	2.9184
p253_005	pair_002	breathy	kokoro	231.1924	211.1685	2.0171	8.1810
p301_005	pair_002	modal	chatterbox	183.1473	184.8020	1.3848	2.9694
p301_005	pair_002	modal	xtts	183.1473	190.8633	1.4155	23.4059
p301_005	pair_002	modal	kokoro	183.1473	211.1685	3.2572	27.5341
p264_006	pair_003	breathy	chatterbox	189.2594	192.2120	0.8281	13.0714
p264_006	pair_003	breathy	xtts	189.2594	189.1510	2.2423	21.3391
p264_006	pair_003	breathy	kokoro	189.2594	202.8054	2.1248	26.0813
p282_006	pair_003	modal	chatterbox	203.9501	204.4833	2.5588	15.9130
p282_006	pair_003	modal	xtts	203.9501	204.0244	2.5577	7.0628
p282_006	pair_003	modal	kokoro	203.9501	202.8054	1.4040	4.8771

Output Files

File	Description
features/features.csv	24 rows, one per audio file with all metrics
features/contours/	24 F0 contour .npy files for DTW analysis
results/gate_check.json	Gate decision and effect sizes per metric
results/per_sample_preservation.csv	Output-reference distances per sample
results/paired_contrast_preservation.csv	Contrast retention by model and pair
results/model_rankings.csv	Model-level summary sorted by score
results/report.html	This report

Caveats

- F0 estimation is sensitive to microphone, noise, and phonetic content.
 - Speaking rate estimation via RMS peak counting is a proxy, not ground truth.
 - DTW alignment may be affected by durational differences between reference and output.
 - Kokoro does not support voice cloning — its outputs reflect the default voice (af_bella), not the reference speaker.
 - **Fragile contrast ratios near zero:** Contrast retention ratios use reference contrast as denominator. When reference contrast is near zero, the ratio becomes unstable. Pair 001 f0_range has reference contrast = 1.52 Hz — effectively zero — producing a ratio of -26.56 for Chatterbox. Clamping to [0, 1] neutralizes the damage, but this pair contributes no f0_range signal to contrast retention scores. This is expected: VCTK pairs were selected for breathiness contrast, not F0 range contrast. Some metrics will have near-zero reference contrasts by coincidence.
 - Single sentence per pair limits generalization to broader prosodic variation.
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Dataset

VCTK Corpus 0.92: <https://datashare.ed.ac.uk/handle/10283/3443>